

Skills Summary

Taylor Polynomials and Remainder Bounds

Use this reference while completing your worksheet or when studying.

A Taylor polynomial answers “what value?”; a remainder bound answers “how wrong could that be?” — and the second is what an exam grades.

1. Building a Taylor polynomial from derivative values

Rule: $T_n(x) = f(c) + f'(c)(x - c) + \frac{f''(c)}{2!}(x - c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x - c)^n$.

The k th derivative *value* is the numerator, $k!$ the denominator, and the power of $(x - c)$ matches k ; Maclaurin is the case $c = 0$.

(!) The factorial is what forces $T_n^{(k)}(c) = f^{(k)}(c)$ — drop it and the polynomial no longer matches the function at all.

Example: $f(1) = 2$, $f'(1) = 6$, $f''(1) = -4$. Approximate $f(1.5)$ with T_2 about $c = 1$.

Step 1. Only three derivative values are given, so T_2 is the best available and every term is built on the increment $x - 1$.

Step 2. $T_2(x) = 2 + 6(x - 1) - \frac{4}{2}(x - 1)^2$, and at $x = 1.5$ the increment is 0.5: $2 + 3 - 2(0.25)$.
4.5

Try it: $g(3) = 1$, $g'(3) = -2$, $g''(3) = 6$. Approximate $g(3.5)$ with T_2 about $c = 3$.

check

2. Building one from a known Maclaurin series

Memorise four: $e^u = \sum \frac{u^n}{n!}$; $\sin u = u - \frac{u^3}{3!} + \frac{u^5}{5!} - \cdots$; $\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - \cdots$;
 $\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots$

Then substitute $u = -x^2$, $u = 2x$, $u = x^3$, whatever the problem needs, and collect up to the degree asked for — differentiating e^{-x^2} six times gives the same answer for ten times the work.

(!) Substitution changes which *degree* each term reaches: with $u = -x^2$, the $n = 3$ term of e^u is already degree 6.

Example: Use the degree-4 Maclaurin polynomial for $\cos x$ to approximate $\cos(\frac{1}{2})$.

Step 1. Cosine's series is known, so no differentiating is needed — just stop it at degree 4 and substitute.

Step 2. $1 - \frac{x^2}{2} + \frac{x^4}{24}$ at $x = \frac{1}{2}$ is $1 - \frac{1}{8} + \frac{1}{384} = \frac{384 - 48 + 1}{384}$.

337
384

Try it: Use the degree-4 Maclaurin polynomial for e^{-x} to approximate e^{-1} .

$\frac{8}{e}$ check

3. Bounding the remainder

Lagrange: $|R_n(x)| \leq \frac{M|x-c|^{n+1}}{(n+1)!}$, where $|f^{(n+1)}(z)| \leq M$ for *every* z between c and x .

Both the exponent and the factorial use $n+1$ — using n is the most common error in the topic.

Alternating series: if the series alternates with terms decreasing to 0, the error is smaller than the **first omitted term** — and its sign says whether the estimate is high or low.

Finding M : bound the $(n+1)$ st derivative on the interval — take $M = e^b$ or larger for e^x on $[0, b]$, and $M = 1$ for sin or cos.

Example: $|f'''(z)| \leq 6$ on $[0, 0.3]$. Bound the error when T_2 about $c = 0$ approximates $f(0.3)$.

Step 1. The polynomial has degree $n = 2$, so the bound needs the third derivative and the third power — that is what $n+1$ means.

Step 2. $\frac{6 \cdot |0.3|^3}{3!} = \frac{6 \cdot 27/1000}{6}$.

$$\frac{27}{1000}$$

Try it: $|f^{(4)}(z)| \leq 24$ on $[1, 1.25]$. Bound the error when T_3 about $c = 1$ approximates $f(1.25)$.

$\frac{927}{1}$ check

4. Using a polynomial to settle a limit

Replace each function by its Maclaurin series, cancel what the numerator subtracts away, and divide term by term: **the first surviving term is the limit.**

(!) Keep enough terms — truncating e^x at $1+x$ makes $e^x - 1 - x$ look like exactly 0.

Example: Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$ using series.

Step 1. Subtracting x kills the leading term of $\sin x$, so the limit is decided by the first term that survives.

Step 2. $\sin x - x = -\frac{x^3}{6} + \frac{x^5}{120} - \dots$, so dividing by x^3 leaves $-\frac{1}{6} + \frac{x^2}{120} - \dots$

$$-\frac{1}{6}$$

Try it: Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ using series.

$\frac{2}{1}$ check